The circular-arc Śrī Yantra

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Abstract

The Śrī Yantra is the union of nine maximal triangles, symmetric about a common axis, whose sides meet in a large number of prescribed coincidences. Chiodo showed that the concurrent models admit a straightedge and compass construction, and that the problem is equivalent to a circle–line–point problem of Apollonius. We consider the configuration obtained by replacing each of the twenty-seven sides by a circular arc, keeping every incidence. We give a straightedge and compass construction of it: twelve seed points carrying eighteen free real parameters, followed by forty-five steps, each of which is a circle through three points or the meeting of two circles. Chiodo's concurrency conditions are satisfied identically — each one is a coincidence of nodes of the configuration rather than a numerical near-miss — so every choice of the eighteen parameters for which the construction is defined yields an exact circular-arc Śrī Yantra. We compute the dimension of the family in two independent ways, by counting the parameters of the construction and by computing the rank of the incidence system, and obtain 18 both times. Two subfamilies, singled out by asking that the outer vertices of one of the two grown rings of cells be concyclic, have dimensions 12 and 11; we show, by reducing the question to a finite sequencing problem and deciding it with a clause-learning solver, that neither condition can be realised by a construction of the same kind, so that these subfamilies can be selected but not built. An interactive viewer and three GeoGebra files accompany the paper.

1 Introduction

The Śrī Yantra, or Śrī Cakra, is a diagram of Tantric Hinduism: a central point, the *bindu*; around it an interlocking figure of nine triangles; around that a ring of eight lotus petals, a ring of sixteen, three circles, and the triple square called the *bhūpura*. The nine-triangle figure is the mathematically interesting part, and it is the part treated here; Figure 1 shows one of the figures constructed below.

Drawing it accurately is a genuine problem. The nine triangles are all isosceles and share a vertical axis of symmetry; their sides are required to pass through a large number of common points, and these requirements are not independent. Traditional methods, transmitted by commentators on the *Saundarya Laharī*, satisfy some of the conditions and miss others, and the discrepancies are visible to the eye. The modern literature on the problem is substantial: Bolton and Macleod [1], Kulaichev [7], Fonseca [3], Tahta [12], Rao [9], Huet [6] and Pustynski [8] each treat the constraints and the accuracy with which they can be met, and Gupta [5] surveys the yantras of mathematical interest more broadly.

The question was settled by Chiodo [2], who gave a straightedge and compass construction of the concurrent models and identified the essential step as a circle–line–point

*Claude Opus 5, model string `claude-opus-5`, run as an agent with file, shell and browser access at maximum reasoning effort. Work carried out in August 2026.

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problem of Apollonius. Chiodo's paper also fixes the notation we use here, and we follow it: the nine maximal triangles are $t_1, \ldots, t_9$, numbered by the height of their bases from the top, with t_i pointing downwards exactly when $i \leq 5$; the *base point* of an isosceles triangle is the midpoint of its base; and its *legs* are the two sides exchanged by the reflection.

Why arcs

Straight sides are not the only tradition. The *Gaurīyāmala Tantra*, quoted by Gupta [5, pp. 181–182], names four architectural forms (*prastāra*) of the figure: *bhū*, the plane form, in which the whole diagram lies in a horizontal plane; *kūrma*, likened to the back of a tortoise, in which the triangular complex is drawn on a spherical surface using spherical triangles; *meru*, in which the enclosures lie in horizontal planes at different heights, after the mythical mountain; and *padma*, the lotus, of which Gupta records only that it does not seem to be popular. The present paper, like Chiodo's, treats the *bhū* form.

The two described non-plane forms differ in kind. In *kūrma* the object remains a surface figure: it lies on a 2-manifold, with its sides geodesics of that manifold, which is the setting of the second question of Section 9. In *meru* the enclosures occupy horizontal planes at different heights, so the figure does not lie on one surface at all, and is three-dimensional in a stronger sense. Rao [9] gives a formulation of the *kūrma* form and optimises it numerically over a range of cap depths, parametrised by an altitude h running from 0 to $\pi/2$; he reports the global optimum near $h = \pi/2$, which is the limit in which the figure occupies a vanishing cap and the plane form is recovered.

The identification of *kūrma* with the sphere deserves a caveat. The name and the image are those of a tortoise's carapace, which is domed but not spherical; Gupta passes from the one to the other within a single sentence, and Rao's formulation is spherical throughout. If the identification is a later convenience, then the surface sought in the second question of Section 9 is not merely a mathematical generalisation of the spherical case: it may be nearer to what the classification named.

A physical instance is the Śrī Cakra carved on the crown of the śivaliṅga at the Chintamaniśvara temple, Anegundi, Koppal district, Karnataka.[1] The crown is a shallow dome rather than a hemisphere. Its innermost triangles are close to straight and the bowing grows outwards, as for a figure laid on a gently curved surface; the result is markedly less distorted than the deeper spherical optima of [9, Fig. 5].

The configuration studied below is not the *kūrma* variant — our arcs lie in the plane, not on a cap — but the two share the thought that the sides of the yantra need not be straight, and that what matters is the incidence structure rather than the straightness. For a shallow cap the two are close: a geodesic there is a short arc near the pole, its orthographic projection is a short piece of an ellipse, and a short elliptical arc is approximated by a circular one to high accuracy.

There is also a purely mathematical reason. The straight figure is rigid in a strong sense: Chiodo's conditions leave a family of small dimension, and the concurrences are hard to satisfy because there is little room to move. Allowing each side to bow — one extra parameter per side, the curvature — enlarges the family considerably, and the interesting question becomes what the incidence structure looks like when there is room to spare. It turns out to look very simple: the conditions can be met not approximately but *identically*, by a construction in which each required coincidence is a single node used twice.

[1] Photographs of the liṅga, from above and in profile, are collected in the spherical gallery of [11].

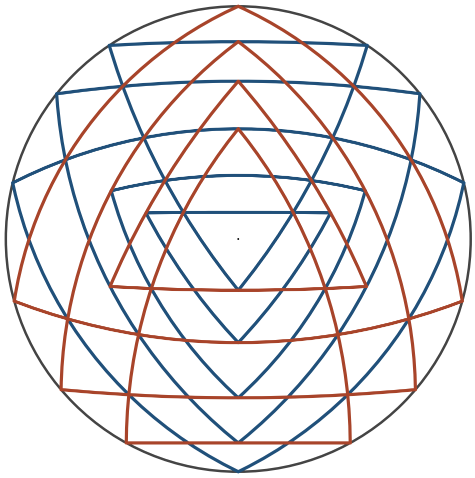

Figure 1: A circular-arc Śrī Yantra: the union of nine circular-arc triangles, five pointing down and four up, all symmetric about the vertical axis. The outer circle is the rim; the bindu is the point at the centre. This is the figure carried by the file `sri-yantra-arcs-free.ggb`.

What is proved

Let Γ denote the incidence structure of the nine-triangle figure: nine triangles, twenty-seven sides, and the sixty-nine nodes at which sides are required to meet. We prove:

- There is a straightedge and compass construction Φ taking twelve seed points, carrying 18 free real parameters, to a realisation of Γ by circular arcs (Section 4). Every step is a circle through three points, or the meeting of two circles, of a circle with the axis, or of a circle with the rim.
- Every figure in the image of Φ satisfies Chiodo's conditions (i), (ii) and (iii) exactly, and does so combinatorially: each of the seven conditions of type (ii) and each of the twelve of type (iii) holds because two objects of the configuration share a node (Theorem 5.1).
- The family has dimension 18. This is the count of free parameters in the construction; it is confirmed independently by the rank of the incidence system, which has 114 equations of rank 88 in 106 unknowns (Theorem 6.1).
- Outside the nine triangles the figure grows two rings of cells. Asking that the outer vertices of the first ring be concyclic is 6 independent conditions and leaves a subfamily of dimension 12; asking it of the second ring is 7 conditions and leaves dimension 11 (Theorem 7.2).
- Neither ring condition can be built into a construction of the kind used here. The question reduces to a sequencing problem, finite but far beyond enumeration, and a clause-learning solver returns unsatisfiable for both rings (Proposition 7.3). The ring-held figures can be selected from the family but not constructed.

Section 8 describes the three GeoGebra files that accompany the paper, one for a distinguished member of each family.

2 The configuration

2.1 Chiodo's conditions

We recall the conditions in the form given in [2, pp. 377–378]. Let $t_1, \ldots, t_9$ be the nine maximal triangles, numbered by the height of their bases from the top, so that t_i points downwards for $i \leq 5$ and upwards for $i \geq 6$. Write $\mathrm{apex}(t)$ for the vertex opposite the base and $\mathrm{bp}(t)$ for the base point.

(i) t_3 and t_7 are inscribed in one and the same circle.

(ii) $\mathrm{apex}(t') = \mathrm{bp}(t'')$ for each of the seven ordered pairs

$$(t', t'') \in \{(t_8, t_1), (t_6, t_2), (t_9, t_3), (t_1, t_6), (t_5, t_7), (t_4, t_8), (t_2, t_9)\}.$$

(iii) On each side of the axis, a leg of the downward triangle t', a leg of the upward triangle t''' and the base of t'' have exactly one point in common, for each of the twelve triples

$$\begin{aligned}(t', t'', t''') \in \{&(t_1, t_2, t_7), (t_2, t_3, t_7), (t_1, t_3, t_8), (t_1, t_4, t_6), (t_1, t_5, t_9), (t_4, t_6, t_9),\\ &(t_2, t_7, t_9), (t_3, t_7, t_8), (t_3, t_8, t_9), (t_4, t_4, t_8), (t_5, t_5, t_6), (t_2, t_6, t_6)\}.\end{aligned}$$

Following [2] we write these as (i), $((\mathrm{ii}); t_i, t_j)$ and $((\mathrm{iii}); t_i, t_j, t_k)$. Conditions (ii) and (iii) are 7 and 12 statements respectively; by the mirror symmetry each statement of type (iii) is one condition, not two.

2.2 Nodes and sides

It is convenient to record the combinatorics once and for all. Each triangle contributes three sides — a base and two legs — so there are 27 sides. The points at which sides are required to meet, together with the corners of the triangles, are the *nodes*; there are 69 of them. The reflection in the axis acts on all of this: it fixes 9 nodes (those on the axis) and exchanges the remaining 60 in 30 pairs; it fixes 9 sides (the bases, which are symmetric) and exchanges the remaining 18 legs in 9 pairs.

Fourteen nodes lie on the rim: the two poles, which are on the axis, and six mirror pairs. In the notation of Section 2 these fourteen include the six corners of t_3 and t_7, which is condition (i) in its sharpest form — in our construction the common circle of (i) is the rim itself.

Figure 2 shows the nine triangles one at a time against a ghost of the whole figure. It is worth looking at for a moment: the nine are of markedly different sizes and eccentricities, and the numbering by base height puts them in an order that has nothing to do with size.

3 Circular-arc Śrī Yantras

3.1 Generalised circles

Throughout, a *generalised circle* is the zero set of

$$\gamma(x, y) \;=\; a\,(x^2 + y^2) \;+\; b\,x \;+\; c\,y \;+\; d, \qquad (a, b, c, d) \in \mathbb{R}^4 \setminus \{0\}, \tag{1}$$

taken up to scale. For $a \neq 0$ this is the circle with centre $(-b/2a,\, -c/2a)$ and radius $\sqrt{b^2 + c^2 - 4ad}\,/\,2|a|$; for $a = 0$ it is a line. Working with (1) rather than with centres and

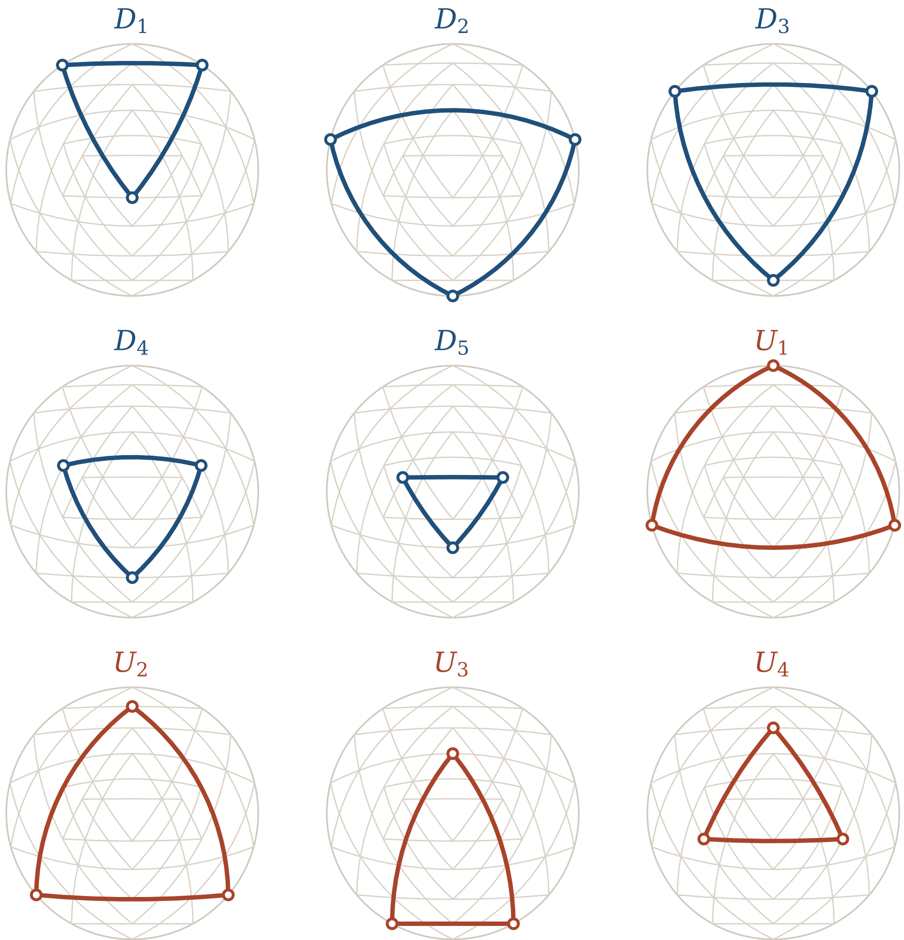

Figure 2: The nine maximal triangles of the same figure, one at a time, in the labelling of the accompanying software (D for downward, U for upward). In Chiodo's numbering by base height these are, in order, $t_1 = D_1$, $t_2 = D_3$, $t_3 = D_2$, $t_4 = D_4$, $t_5 = D_5$, $t_6 = U_4$, $t_7 = U_1$, $t_8 = U_2$, $t_9 = U_3$, and the same correspondence holds at every member of the arc family considered here. It is not universal: the numbering follows the base heights, which move with the parameters, and on the straight figure of Section 6 the first three are instead $t_1 = D_3$, $t_2 = D_2$, $t_3 = D_1$. Open circles mark the three corners.

radii has two advantages: lines are not a special case that has to be handled separately, and the condition that a point lie on the curve is linear in (a, b, c, d), so the generalised circle through three given points is the null vector of a 3×4 matrix,

$$\begin{pmatrix} |P|^2 & P_x & P_y & 1 \\ |Q|^2 & Q_x & Q_y & 1 \\ |R|^2 & R_x & R_y & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = 0, \tag{2}$$

which is a determinant computation and never requires a case distinction. A straight side is the case $a = 0$ and is reached continuously, so the straight Śrī Yantra is a limit of circular-arc ones rather than a separate object.

Definition 3.1. A *circular-arc Śrī Yantra* is a realisation of the incidence structure Γ in which each of the twenty-seven sides is an arc of a generalised circle, the whole figure is symmetric about a vertical axis, and every node required to lie on a side does lie on it.

We stress that Definition 3.1 imposes nothing beyond the incidences and the symmetry. In particular the two grown rings discussed in Section 7 are not constrained, and t_3 and t_7 are not required to be inscribed in a common circle — though in the construction of Section 4 they always are, and in the rim.

4 The construction

4.1 The seeds

The construction begins with twelve points, listed in Table 1 and shown on the disc in Figure 3. Eight of them are free in the plane and contribute two numbers each; one is constrained to the rim and contributes its angle; one is constrained to the axis and contributes its height; two are pinned at the poles $(0, \pm 1)$ and contribute nothing. The total is 18 free real numbers. Each seed off the axis is accompanied by its mirror image, which costs nothing further.

seed	constrained to	numbers	value in the free Default
P_{10}	free in the plane	2	$(-0.273920,\ 0.844094)$
P_5	free in the plane	2	$(-0.309849,\ -0.041519)$
P_{12}	on the rim	1	$\theta = 0.982535$
P_8	free in the plane	2	$(0.236255,\ 0.114118)$
P_{46}	on the axis	1	$y = 0.473624$
P_{32}	free in the plane	2	$(0.743697,\ 0.340431)$
P_{68}	pinned at a pole	0	$(0,\ +1)$
P_{60}	free in the plane	2	$(0.215835,\ -0.876275)$
P_{23}	free in the plane	2	$(-0.551486,\ -0.204177)$
P_2	free in the plane	2	$(0.190141,\ -0.218411)$
P_{62}	free in the plane	2	$(-0.467600,\ -0.021710)$
P_{55}	pinned at a pole	0	$(0,\ -1)$
total		18	

Table 1: The twelve seed points and the eighteen numbers they carry. The values are those of the distinguished member described in Section 8.

4.2 The five moves

From the seeds the figure is completed in forty-five steps, each of one of five kinds (Table 2 and Figure 4). Two produce generalised circles and three produce points:

(B) Given a node P off the axis and a further node Q, take the generalised circle through P, its mirror image $\bar{P}$ and Q. Such a circle is automatically symmetric about the axis, and this is how the nine bases are produced.

(C) Given three nodes, take the generalised circle through them, by (2).

(X) Given two generalised circles, take one of their two points of intersection.

(A) Given a generalised circle, take one of its two meetings with the axis.

(R) Given a generalised circle, take one of its two meetings with the rim.

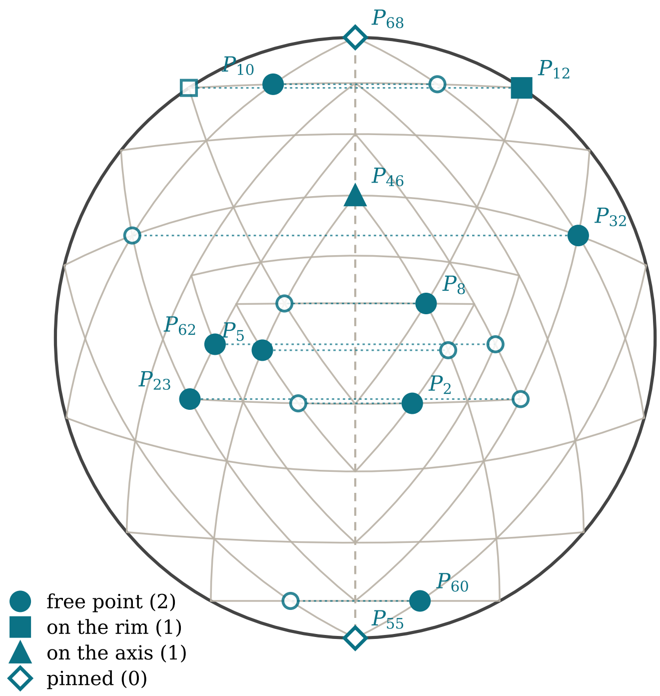

Figure 3: The twelve seeds on the disc, with the finished figure behind them in grey. Filled markers are the seeds proper; open markers are their mirror images, joined to them by dotted lines. The two diamonds at the poles are pinned and carry no parameter.

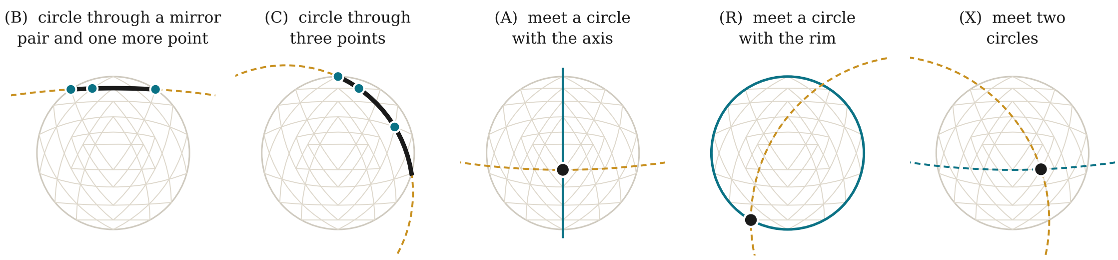

Figure 4: The five moves, each shown at the step of the construction where it first occurs. Dashed curves are the generalised circles being used, the heavy black arc or dot is what the step produces, and the figure is behind in grey.

move	what it produces	times used
(B)	the generalised circle through a mirror pair $P, \bar{P}$ and one further point	9
(C)	the generalised circle through three points	9
(X)	the meeting of two generalised circles	16
(A)	the meeting of a generalised circle with the axis	6
(R)	the meeting of a generalised circle with the rim	5
total steps		45

Table 2: The five moves and how often each is used.

Moves (X), (A) and (R) each have two candidate outcomes, and the construction must say which. This is the one place where the recipe carries data beyond the eighteen parameters: each such step records an approximate location, and the branch taken is the one nearer to it. The choice is discrete, so it is locally constant in the parameters; the

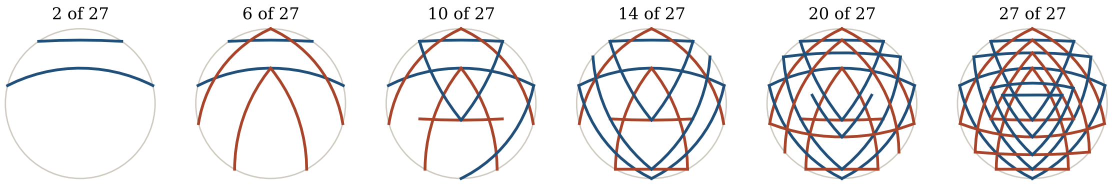

Figure 5: The construction unfolding. Each panel shows the arcs available after a further block of steps; downward triangles in blue, upward in red.

construction is therefore a well-defined analytic map on each connected component of the set of parameters for which no two candidate branches collide. The locus on which two branches coincide is precisely where the construction degenerates.

Remark 4.1. Every move is a straightedge and compass operation. (B) and (C) are the circumscribed circle of three points, which is the classical construction by perpendicular bisectors; (X), (A) and (R) are intersections of a circle with a circle or a line. Nothing here needs the Apollonius step that Chiodo [2] requires for the straight figure, and nothing needs a conic. This is not because the arc problem is intrinsically simpler, but because the additional curvature parameters afford sufficient freedom to place the objects directly, so that no auxiliary tangency problem arises.

4.3 The order of construction

Figure 5 shows the twenty-seven sides appearing in the order the recipe produces them. The bases of the largest triangles come first, because they can be drawn from the seeds alone; the smaller triangles come last, because their corners are intersections of arcs that do not exist until the larger ones are in place.

5 Exactness

The point of a construction, as opposed to a numerical fit, is that the conditions hold because of how the figure is made rather than because a residual has been driven small. For the circular-arc yantra this is true in a strong form: Chiodo's conditions hold not merely to machine precision but *identically*, in the sense that the two objects a condition asserts to be equal are literally the same node of the configuration.

Theorem 5.1. *Let F be any figure in the image of the construction Φ of Section 4. Then F satisfies conditions* (i), (ii) *and* (iii) *of Section 2. Moreover:*

- *(a) each of the seven conditions of type* (ii) *holds because* $\mathrm{apex}(t')$ *and* $\mathrm{bp}(t'')$ *are the same node of* Γ*;*
- *(b) each of the twelve conditions of type* (iii) *holds because a single node of* Γ *lies on all three of the arcs concerned;*
- *(c) condition* (i) *holds with the common circle equal to the rim: the six corners of* t_3 *and* t_7 *are among the fourteen nodes constrained to the unit circle.*

Proof. All three statements are about the incidence structure Γ, not about the particular parameters, so it suffices to exhibit the relevant nodes; they are recorded in Table 3. For (a), in each of the seven pairs the node listed is by definition the apex of t' — the node where the two legs of t' meet — and is also the middle node of the base chain of t'', which is the node at which that base crosses the axis. Since the base is symmetric about the axis, its crossing with the axis is its base point, so $\mathrm{apex}(t') = \mathrm{bp}(t'')$ as required. For (b), in each of the twelve triples the node listed belongs to the node chain of a leg of t', of a leg of t''' and of the base of t''; since Φ places every node of a chain on the corresponding arc, the three arcs share that node. Uniqueness of the common point is the statement that no two of the three arcs coincide, which holds on the open set where the construction is defined. For (c), the corners in question are the two ends of the base of t_3, its apex, and the same three for t_7; the construction places all six on the rim, two of them being the pinned poles. □

Figure 6 shows where in the figure the incidences fall.

condition	node	condition	node
$((\mathrm{ii}); t_8, t_1)$	67	$((\mathrm{iii}); t_1, t_2, t_7)$	13
$((\mathrm{ii}); t_6, t_2)$	61	$((\mathrm{iii}); t_2, t_3, t_7)$	32
$((\mathrm{ii}); t_9, t_3)$	46	$((\mathrm{iii}); t_1, t_3, t_8)$	14
$((\mathrm{ii}); t_1, t_6)$	16	$((\mathrm{iii}); t_1, t_4, t_6)$	15
$((\mathrm{ii}); t_5, t_7)$	3	$((\mathrm{iii}); t_1, t_5, t_9)$	8
$((\mathrm{ii}); t_4, t_8)$	27	$((\mathrm{iii}); t_4, t_6, t_9)$	21
$((\mathrm{ii}); t_2, t_9)$	36	$((\mathrm{iii}); t_2, t_7, t_9)$	34
		$((\mathrm{iii}); t_3, t_7, t_8)$	48
		$((\mathrm{iii}); t_3, t_8, t_9)$	59
		$((\mathrm{iii}); t_4, t_4, t_8)$	24
		$((\mathrm{iii}); t_5, t_5, t_6)$	0
		$((\mathrm{iii}); t_2, t_6, t_6)$	20

Table 3: The node of Γ that makes each of Chiodo's conditions hold. The numbering of nodes is that of the accompanying files. Because these are statements about Γ, the same table serves every member of every family considered here.

Remark 5.2. Theorem 5.1 does not assert that the circular-arc problem is trivial. Rather, the difficulty is relocated: the conditions are free once the incidence structure is realised at all, and the whole content of the problem is in realising it — that is, in the forty-five steps, which have to be arranged so that every node lands on every arc it is supposed to. The straight case is hard because that arrangement is over-determined; the arc case is tractable because it is not.

6 The dimension of the family

Theorem 6.1. *The family of circular-arc Śrī Yantras produced by Φ has dimension* 18.

The number can be obtained in two ways which share no step, and they agree.

First count: the construction

Table 1 lists the parameters: eight free points at two numbers each, one angle on the rim, one height on the axis, and two pinned poles, giving $16 + 1 + 1 + 0 = 18$. Every later step is determined by what precedes it, so Φ is a map from an open subset of $\mathbb{R}^{18}$ to configurations. It is injective on the set where it is defined, since the seeds are recovered from the figure as particular nodes.

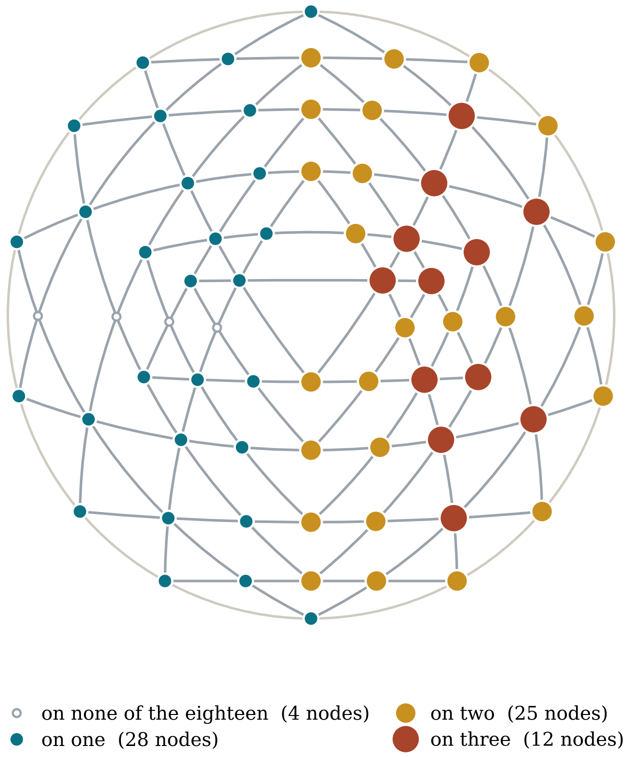

Figure 6: Where the incidences sit. Each node is drawn at a size proportional to the number of the eighteen mirror-representative sides it is required to lie on. Exactly twelve nodes carry three incidences each, and they are exactly the twelve nodes of the right-hand column of Table 3: the points at which Chiodo's condition (iii) is realised. Their mirror images carry no incidences here, because only one side of each mirror pair is represented in the system.

Second count: the incidence system

Alternatively, forget the construction and impose the incidences directly. Parametrise a symmetric configuration by the free coordinates of its nodes and the coefficients of its arcs, modulo the symmetry. Counting as in Section 2: seven nodes on the axis contribute one number each, six mirror pairs on the rim contribute one angle each, and the remaining twenty-four mirror pairs contribute two numbers each, giving $7+6+48=61$ node parameters. Nine of the sides are symmetric bases, each of which is a generalised circle with a horizontal tangent direction imposed and so carries two parameters; the other eighteen fall into nine mirror pairs, one representative of each carrying the three parameters of a generalised circle up to scale, giving $18+27=45$ arc parameters. The total is 106.

The incidences to be imposed are, for each of the eighteen mirror-representative sides, that each node of its chain lie on it. That is 114 scalar equations. Their Jacobian at the distinguished figure of Section 8 has rank 88, so 26 of the 114 are dependent, and the solution set is smooth of dimension $106-88=18$ there.

	count	
node parameters	61	7 axis nodes, 6 rim pairs, 24 free pairs
arc parameters	45	9 symmetric bases, 9 leg pairs
unknowns	**106**	
incidence equations	114	on the 18 mirror-representative sides
rank of the system	88	computed at the Default figure
dimension	**18**	= 106 − 88
free numbers in the construction	**18**	counted from Table 1

Table 4: The two counts. The last line is the count from the construction; the line above it is the count from the incidence system. They agree.

Remark 6.2. The agreement is a check on both. The construction could have been redundant, producing a family of dimension less than 18 because two of its parameters moved the figure the same way; the incidence system could have been miscounted, or evaluated at a non-generic point. Neither is the case. We computed the rank at several members of the family and obtained 88 each time; we have not proved that the rank is 88 at every point, and there is no reason to expect it to be — the rank can only drop on the locus where the figure degenerates, and it is exactly there that the construction fails as well. The six members tested were the distinguished figure of each of the three families and a perturbation of each.

7 The grown rings, and two subfamilies

Outside the nine triangles the arcs continue to cross one another, and the crossings bound two further rings of cells, of fourteen cells each. We call their outer vertices *tips*.

These rings should not be confused with the classical enclosures. The traditional description — the *Rudrayāmala Tantra*, as reported by Gupta [5] and by Chiodo [2, p. 379] — speaks of the bindu, a central triangle, and enclosures of 8, 10, 10 and 14 triangles, the outermost being the *caturdaśāra*. Those lie *inside* the nine triangles, as parts of the subdivision the triangles cut out of one another. The rings considered here are what the twenty-seven arcs bound when they are continued *outside* the nine.

Remark 7.1. That region is not empty in the traditional figure either. Between the yantra's circle and the square *bhūpura* the painted and engraved Śrī Yantras carry two rings of lotus petals: the *aṣṭadala padma* of eight petals within, and the *ṣoḍaśadala padma* of sixteen without. These are two of the nine *āvaraṇas* into which the figure is traditionally divided, and in the received drawings they are ornament: their number and their placing are given, but nothing in the nine triangles produces them.

The grown rings occupy exactly that annulus, and they are produced by the figure — they are what the twenty-seven arcs cut out of the plane immediately beyond the triangles. Figure 7 sets the two side by side. We do not suggest a historical derivation, and the counts do not agree: the grown rings have fourteen cells each, against eight and sixteen petals. But a ring of pointed cells standing on a circle and reaching to another is the shape a lotus ring has, and the construction supplies one without being asked. The classical requirement that each āvaraṇa close on a circle is, in this reading, the same requirement as concyclicity of a ring's tips, which is the condition of Theorem 7.2.

For completeness we record the interior subdivision. The nine triangles cut the disc into 74 atomic regions: the central one, and then shells of $3, 8, 8, 10, 10, 10, 10$ and 14

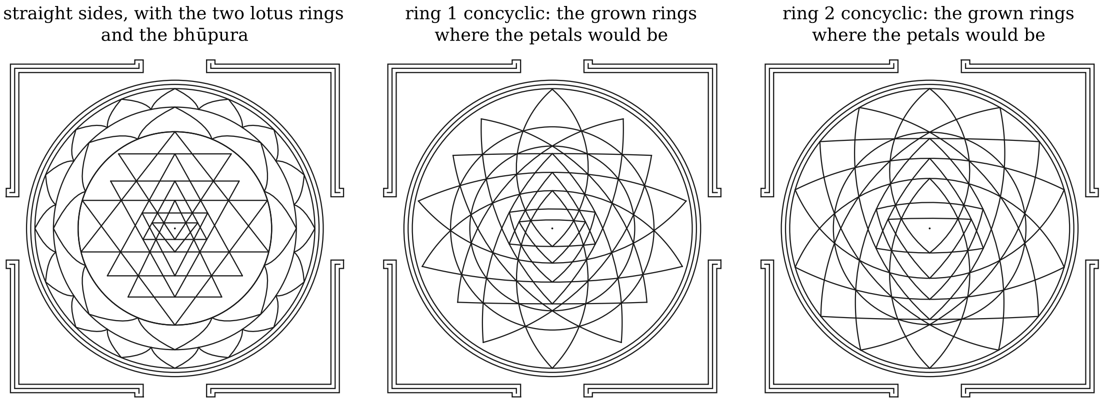

Figure 7: Left: the figure as traditionally drawn, with straight sides and the eight- and sixteen-petalled lotus rings. Middle and right: the two ring-held members of the circular-arc family, without petals, the grown rings occupying the same annulus. In the middle panel the circle carrying the first ring's tips is drawn explicitly, that ring lying interior to the second; in the right-hand panel the second ring's tips are of maximal radius, so the innermost frame circle already passes through them. Each panel carries the same frame, in an open form of the *bhūpura*: three concentric circles and a band broken at the middle of each side, its four corners closing on themselves. The innermost circle passes through the outermost points of each figure — the outer petal tips on the left, the outer ring tips on the other two. All strokes are of one weight.

reading outwards. This is finer than the classical count of 43, which groups some of these regions together; the decomposition is the same for the straight figure and for every member of the arc family, since it depends only on Γ. In a general member of the family the tips of a ring are not concyclic, and they are not nearly concyclic either: at the distinguished figure of the free family the radii of the first ring's tips span 22.3% of their mean and those of the second ring 135.8%. Concyclicity is a real restriction, not a small correction.

Asking for them to be concyclic is a natural condition, and it is the arc analogue of the traditional requirement that the enclosures close on circles.

Theorem 7.2. *Within the* 18*-dimensional family:*

(a) the condition that the fourteen tips of the first ring be concyclic has rank 6*, and cuts out a subfamily of dimension* 12*;*

(b) the condition that the fourteen tips of the second ring be concyclic has rank 7*, and cuts out a subfamily of dimension* 11*.*

Figure 8 shows the three families side by side.

The ranks are computed from the Jacobian, in the eighteen seed parameters, of the fourteen residuals "radius of tip i minus the mean radius". That these ranks are 6 and 7 rather than 13 is mirror symmetry doing most of the work: the fourteen tips fall into mirror classes, and within a class the radii are equal automatically.

7.1 The ring conditions cannot be constructed

Theorem 7.2 locates the two subfamilies but says nothing about how a member of one might be drawn. For constructions of the kind used in Section 4 the answer is negative, and provably so.

Proposition 7.3. *Consider constructions of the following form. Each circle, each node and each ring tip is produced exactly once, in some order. A circle is produced at a moment when exactly as many of the nodes on it already exist as determine it — three in general, two for the axis-symmetric bases. A node is produced either at a moment when exactly as many of the circles through it exist as determine it, or from no circle at all, in which case it is a seed and costs its parameters. A ring tip is produced after exactly one of the two circles meeting there and before the other, so that the first cuts it out of the ring's circle and the second is drawn through it; this is what would place the ring on a circle by construction rather than by fitting. Require finally that the seeds carry the parameters permitted by Theorem 7.2: eleven for the first ring and ten for the second, which with the ring's own radius give* 12 *and* 11*.*

Then no such order exists, for either ring.

Proof. The requirements are conditions on the relative order of finitely many objects — for the first ring, 18 circle orbits, 39 node orbits and 7 tip orbits, the orbits being those of the mirror symmetry. Coverage of the incidences is then forced rather than chosen: a point of a circle existing before the circle and not among its determining points is an incidence that nothing can subsequently cover, while a point produced later is produced from the circle. The question is therefore a finite constraint-satisfaction problem in the positions of the objects within the order.

It cannot be settled by enumeration: the number of orders is about 3×10^{64}, and a breadth-first sweep is already 3.7 million states wide at level 6 of 64. Posed for a conflict-driven clause-learning solver it is nevertheless decided. We used `z3` [10], splitting the search by cube-and-conquer in both cases: the sets of objects reachable after a fixed number of steps partition the search into disjoint parts, each refuted independently. For the first ring the partition after four steps has 84,211 parts. The solver returned *unsatisfiable* for both rings. The scripts are in the repository named in Section 8. □

Remark 7.4. The second ring differs from the first in a way that might have been expected to help. Its fourteen tips fall into eight orbits rather than seven, and two of these lie on the axis, being the crossings of a leg with its own mirror image. An axis tip is available before any circle exists, since it is where the ring's circle meets the axis, and the leg concerned may then be built through it. The advantage is not sufficient.

Remark 7.5. Proposition 7.3 does not assert that the ring-held figures fail to exist. Theorem 7.2 exhibits families of dimensions 12 and 11, and the accompanying files carry a member of each. It asserts that such a figure cannot be *built*, only *selected*: the concyclicity is a condition on the eighteen parameters, satisfied by solving, and not a consequence of any straightedge and compass procedure of the form described. In this the two subfamilies differ from the family as a whole, every member of which is constructible by Section 4. It is also what the reader will observe on dragging a seed in either of the two ring files: the incidences survive and the concyclicity does not.

The result is relative to the class of constructions specified. Permitting an object to be drawn more than once, or auxiliary objects outside the figure, or steps beyond the five of Section 4, would require the question to be posed afresh.

Remark 7.6. The ranks quoted are computed *on* the locus in question, and this matters. Evaluated instead at a general member of the 18-dimensional family, where neither ring is concyclic, the first ring's system has rank 7 and the second's rank 8. The ranks therefore drop by one on each of the two loci: these subfamilies are precisely where the conditions

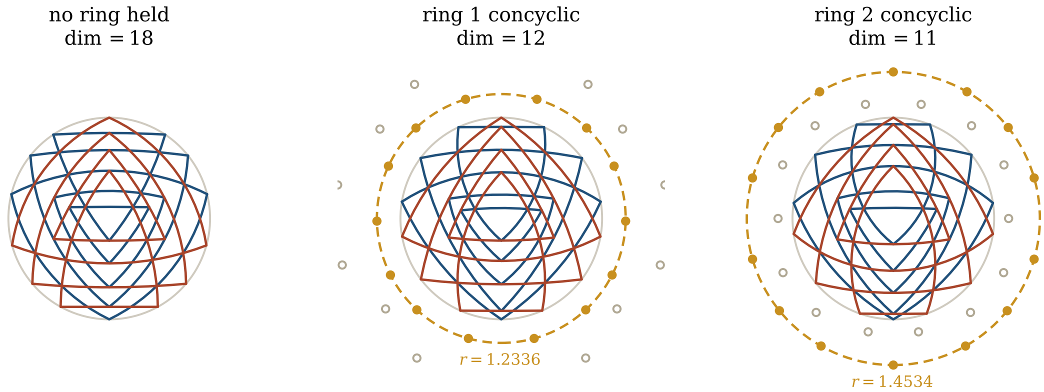

Figure 8: The three families. Left: no ring held, the general case, dimension 18. Middle and right: the first and the second grown ring required to be concyclic, dimensions 12 and 11. Filled dots are the tips of the held ring and the dashed circle is the circle they lie on; open dots are the tips of the other ring, which are not concyclic.

become degenerate. The dimensions 12 and 11 quoted in Theorem 7.2 are the ones that match the construction, in the sense that the tangent space computed from the seeds at a ring-held figure has exactly that dimension.

Remark 7.7. The two conditions are not nested: holding the first ring does not hold the second, nor conversely. Whether a circular-arc Śrī Yantra can satisfy both simultaneously is not addressed here. We note only that a rank computed at a figure satisfying neither condition would not settle the question, for the reason given in Remark 7.6: the ranks of these systems fall on the loci where the conditions hold.

8 The accompanying GeoGebra files

An interactive viewer accompanies this paper, together with the three GeoGebra files described below. Both are available at

https://github.com/michaelkopp/sriyantra

The viewer draws any member of the three families from its eighteen parameters, in the principal directions of Section 6 or in the seed coordinates themselves, and exports the figure as SVG or PNG.

The three GeoGebra files are:

file	family	dimension
`sri-yantra-arcs-free.ggb`	no ring held	18
`sri-yantra-arcs-ring1.ggb`	first ring concyclic	12
`sri-yantra-arcs-ring2.ggb`	second ring concyclic	11

Each carries one distinguished member of its family, the one used for the figures above and for the numerical statements in the text. The files are not static drawings. Each contains the construction itself, written out as GeoGebra [4] commands and evaluated when the file is opened: the twelve seeds as free or constrained points, then the forty-five steps as `Circle` through three points and `Intersect` of two objects, then the twenty-seven sides as circular arcs. The two files for the ring-held families carry the grown rings as well: each ring cell is drawn from the same twenty-seven circles, continued outside the yantra, its tip being where two of them cross. Ring 1's cells rest on nodes of the yantra and ring 2's on the tips of ring 1, so the second ring is built on the first. Those two files also carry the circle the held ring sits on, drawn in red, so that the condition defining the family is visible rather than merely asserted. Dragging a seed re-derives the whole figure, rings included. The reader who drags one will find that the incidences of Theorem 5.1 survive — they are steps of the construction — while the concyclicity of a ring, in the two files where it holds, does not: it is a condition on the parameters, not a consequence of the construction.

The branch choices of moves (X), (A) and (R) are carried in the files as the nearest-point rule described in Section 4, so that a small drag keeps the intended branch. A large drag can cross a wall where two branches meet, after which the figure will jump; this is a real feature of the construction rather than an artefact of the file.

We verified the files by re-evaluating their command lists in an independent implementation of the same operations — circles through three points by (2), conic intersections by the radical line — and comparing all sixty-nine nodes with the values produced by the construction. The largest disagreement is 2.2×10^{-12}, consistent with the twelve decimal places to which the seed values are written into the files.

Software

The construction, the figures and the numerical statements in this paper were produced with the viewer at https://github.com/michaelkopp/sriyantra, which is distributed with the three GeoGebra files.

9 Concluding remarks

The circular-arc Śrī Yantra is a considerably less rigid object than the straight one. Where Chiodo must solve an Apollonius problem to close the straight figure, the arc figure closes by elementary steps, and the concurrency conditions that make the straight problem delicate are here consequences of the incidence structure alone. The interest moves from whether the figure can be drawn to which of the eighteen-parameter family one should choose, and it is here that the classical conditions — the enclosures closing on circles — return as genuine constraints, cutting the family to twelve and to eleven.

Three questions remain.

1. The straight Śrī Yantra is the locus $a = 0$ for all twenty-seven arcs, which is 27 conditions on 18 parameters and so is not reached inside this family. The straight and arc constructions are therefore not specialisations of one another, although both realise Γ. Is there a construction containing both? We conjecture that conic sides suffice: a conic through five points carries two parameters more than a circle through three, and the straight figure would arise as a degeneration within one connected family. The incidence structure would require re-examination, a conic through five points being a step of a different character.

2. Of the twenty-seven corners of the nine triangles, fourteen lie on the boundary of their union; which fourteen depends only on Γ, not on the realisation. Call a realisation *geodesic* if its twenty-seven sides are geodesics of a Riemannian surface, and let $\nu \leq 14$ count those outer corners lying on the circumscribed circle. The construction of Section 4 attains the maximum $\nu = 14$, with no corner outside, but

is not geodesic. In the Euclidean plane the geodesic realisations are the straight figures, where ν depends on the member: $\nu = 6$ for that of Section 6, the corners of t_3 and t_7 supplied by condition (i), and $\nu = 10$ for Kulaichev's [7], which puts the ends of the bases of t_1 and t_9 on the circle as well. Rao [9] treats the spherical case but optimises against other criteria, so the maximum of ν there is not settled by his figures. Is there a Riemannian surface admitting a geodesic realisation with $\nu = 14$, and is any such realisation aesthetically satisfying?

3. Which member of the family should be selected? This is not, in the end, a mathematical question. The criterion governing the form of the Śrī Yantra has always been in part aesthetic, and while clarity and uniqueness of construction contribute to the quality of a figure, the two need not agree: optimisation of a metric quantity in isolation readily yields figures of little contemplative value. The distinguished members displayed in this article and carried by the accompanying files were selected on aesthetic grounds.

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